ABSTRACT OF THE DISCLOSURE

A mass storage system configured to physically store data using a high-radix encoding scheme, where each storage element (e.g., a memory cell, a magnetic domain) can reliably represent one of N discrete, distinguishable physical states, with N being an integer greater than two. This system comprises a high-radix storage medium (e.g., flash memory, magnetic media, optical media) and a high-radix storage controller. The controller is configured to write abstract symbols directly into the storage medium by setting the physical state of corresponding storage elements and to read data by detecting these states and converting them back into abstract symbols. This architecture fundamentally moves beyond binary data storage, enabling higher data density per physical element and a more direct representation of information for parametric computing systems, reducing the need for binary conversion at the storage interface.